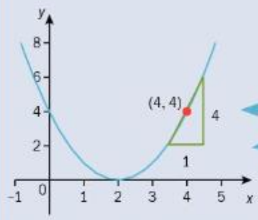
1) Differentiation

a) Gradient of tangent

Example 1

Estimate the gradient of the tangent to the curve $y = (x - 2)^2$ at the point (4, 4).



We can only *estimate* the gradient as it is difficult to draw the tangent accurately.

Hint: Draw the base of the triangle with an integer value.

Gradient of the tangent at (4, 4) $= \frac{4}{1} = 4$

b) Differentiation of polynomials

Example 3

Find the derived function in each case.

a) $f(x) = x^8$

b) $f(x) = \frac{6}{x^4}$

c) $f(x) = 3x$

d) $f(x) = 5$

Continued on the next page

a) $f(x) = x^8$

$f'(x) = 8x^{8-1} = 8x^7$

Use nx^{n-1}.

b) $f(x) = \frac{6}{x^4} = 6x^{-4}$

First write the power as a negative power.

$f'(x) = -24x^{-5}$

$= -\frac{24}{x^5}$

$-4 \times 6 = -24$ and -5 is one less than -4.

c) $f(x) = 3x$

$f'(x) = 3$

$f(x) = 3x^1$, $f'(x) = 3x^{1-1} = 3x^0 = 3 \times 1 = 3$

d) $f(x) = 5 = 5x^0$

$f'(x) = 0$

Note: When we differentiate a constant, c, we always get 0. This makes sense as the graph of $f(x) = c$ is a horizontal straight line.

Example 4

Find $\frac{dy}{dx}$ when $y = 3\sqrt{x}$.

$y = 3\sqrt{x} = 3x^{\frac{1}{2}}$

First write $\sqrt{x}$ as $x^{\frac{1}{2}}$.

$\frac{dy}{dx} = \frac{3}{2}x^{-\frac{1}{2}}$

$= \frac{3}{2\sqrt{x}}$

$x^{-\frac{1}{2}} = \frac{1}{\sqrt{x}}$

Example 5

Find $\frac{dy}{dx}$ when

a) $y = x^5 + 4x^4$ **b)** $y = (2x - 5)^2$ **c)** $y = \frac{3x^6 + 8}{x^2}$.

a) $y = x^5 + 4x^4$

$\frac{dy}{dx} = 5x^4 + 16x^3$ — Differentiate term by term.

b) $y = (2x - 5)^2 = 4x^2 - 20x + 25$ — First expand the brackets.

$\frac{dy}{dx} = 8x - 20$ — When you differentiate a constant it is 0.

c) $y = \frac{3x^6 + 8}{x^2} = \frac{3x^6}{x^2} + \frac{8}{x^2} = 3x^4 + 8x^{-2}$ — First divide each term by x^2.

$\frac{dy}{dx} = 12x^3 - 16x^{-3}$

$= 12x^3 - \frac{16}{x^3}$

c) Chain rule

Example 7

Find $\frac{dy}{du}$ when $y = \sqrt{4 - 3x}$.

Let $u = 4 - 3x$ — Let u = the expression in the square root.

$\frac{du}{dx} = -3$

$y = \sqrt{u} = u^{\frac{1}{2}}$

$\frac{dy}{du} = \frac{1}{2}u^{-\frac{1}{2}}$

$= \frac{1}{2}(4 - 3x)^{-\frac{1}{2}}$ — Now substitute $u = 4 - 3x$.

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = \frac{1}{2}(4 - 3x)^{-\frac{1}{2}} \times -3$ — Apply the chain rule.

$= -\frac{3}{2}(4 - 3x)^{-\frac{1}{2}}$ — Simplify your answer.

$= -\frac{3}{2\sqrt{4 - 3x}}$

Example 8

Find $\frac{dy}{dx}$ when $y = \frac{2}{(3x^2+1)^4}$.

Let $u = 3x^2 + 1$ — Let u = the expression in the bracket.

$\frac{du}{dx} = 6x$

$y = 2u^{-4}$

$\frac{dy}{du} = -8u^{-5}$

$= -8(3x^2+1)^{-5}$ — Now substitute $u = 3x^2 + 1$.

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = -8(3x^2+1)^{-5} \times 6x$ — Apply the chain rule.

$= -48x(3x^2+1)^{-5}$ — Simplify your answer.

$= -\frac{48x}{(3x^2+1)^5}$

d) Gradient of tangent

Example 9

Find the gradient of the tangent to the curve $y = 4x^3 - 2x^2 + 1$ at the point (1, 3).

$\frac{dy}{dx} = 12x^2 - 4x$ — $\frac{dy}{dx}$ gives us the gradient at any point on the curve.

At $x = 1$, $\frac{dy}{dx} = 12(1)^2 - 4(1) = 8$ — We only need to use the x-value.

Gradient of the tangent = 8

Example 10

Find the coordinates of the point on the curve with equation $y = 5 - 3x - 4x^2$ where the gradient is 1.

$\frac{dy}{dx} = -3 - 8x = 1$ — Put $\frac{dy}{dx} = 1$ as the gradient = 1.

$-8x = 4$

$x = -0.5$

When $x = -0.5$,

$y = 5 - 3(-0.5) - 4(-0.5)^2 = 5.5$ — Substitute in the original equation to find y.

Coordinates are $(-0.5, 5.5)$.

Example 11

Find the gradient of the tangent to the curve $y = 4x^2 - 5x + 2$ at each of the points where the curve meets the line $y = 7x - 6$.

The curve meets the line when

$$4x^2 - 5x + 2 = 7x - 6$$

The y-values are equal at the points of intersection.

$$4x^2 - 12x + 8 = 0$$

$$x^2 - 3x + 2 = 0$$

It is easier to factorise if we divide the equation by the common factor of 4.

$$(x - 1)(x - 2) = 0$$

$$x = 1 \text{ or } x = 2$$

$$\frac{dy}{dx} = 8x - 5$$

Differentiate to find an expression for the gradient.

When $x = 1$, the gradient of the tangent $= 8(1) - 5 = 3$.

When $x = 2$, the gradient of the tangent $= 8(2) - 5 = 11$.

e) Second derivative

Example 12

Find $\frac{d^2y}{dx^2}$ when $y = (4x^2 + 1)^2$.

$$y = (4x^2 + 1)(4x^2 + 1) = 16x^4 + 8x^2 + 1$$

First expand the brackets.

$$\frac{dy}{dx} = 64x^3 + 16x$$

Differentiate each of the three terms.

$$\frac{d^2y}{dx^2} = 192x^2 + 16$$

Then differentiate again.

Example 13

Find $\frac{d^2y}{dx^2}$ when $\frac{dy}{dx} = \frac{3}{\sqrt{x}}$.

$$\frac{dy}{dx} = \frac{3}{\sqrt{x}} = 3x^{-\frac{1}{2}}$$

Remember $\frac{3}{\sqrt{x}} = \frac{3}{x^{\frac{1}{2}}} = 3x^{-\frac{1}{2}}$

$$\frac{d^2y}{dx^2} = -\frac{3}{2}x^{-\frac{3}{2}}$$

We only need to differentiate once when given $\frac{dy}{dx}$.

$$= -\frac{3}{2\sqrt{x^3}}$$

$x^{\frac{3}{2}} = \sqrt{x^3}$ or $(\sqrt{x})^3$ or $x\sqrt{x}$

Example 14

Given that $f(x) = 8x^3 - 3x + \frac{4}{x}$, find the value of $f''(-2)$.

$$f'(x) = 24x^2 - 3 - \frac{4}{x^2}$$

Differentiate $f(x)$ to get $f'(x)$.

$$f''(x) = 48x + \frac{8}{x^3}$$

Differentiate $f'(x)$ to get $f''(x)$.

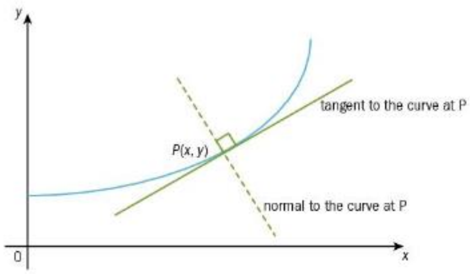
$$f''(-2) = 48(-2) + \frac{8}{(-2)^3}$$

$$= -96 - 1 = -97$$

Substitute $x = 2$ in to $f''(x)$.

f) Equation of tangents and normal

We can use differentiation to find the gradient of the tangent to a curve at a particular point. We will then be able to find the **equation of the tangent** and the **equation of the normal** at this point by using our knowledge of coordinate geometry.



Example 15

Find the equation of the tangent to the curve $y = 3x^3 - 6x^2 + x$ at the point $(1, -2)$.

$y = 3x^3 - 6x^2 + x$

$\frac{dy}{dx} = 9x^2 - 12x + 1$ ← Differentiate to find the gradient of the tangent.

When $x = 1$, $\frac{dy}{dx} = 9(1)^2 - 12(1) + 1 = -2$. ← Substitute to find the gradient at $(1, -2)$.

Tangent goes through $(1, -2)$ with $m = -2$.

$y + 2 = -2(x - 1)$ ← We use $(1, -2)$ and $m = -2$ to find the equation.

$y + 2 = -2x + 2$

Equation of tangent is $y + 2x = 0$.

Example 16

Find the equation of the normal to the curve $y = 4\sqrt{x} + 7$ at the point where $x = 16$.

$y = 4\sqrt{x} + 7 = 4x^{\frac{1}{2}} + 7$

Write the expression using indices.

$\frac{dy}{dx} = 2x^{-\frac{1}{2}} = \frac{2}{\sqrt{x}}$

When $x = 16$, $y = 4\sqrt{16} + 7 = 23$

Substitute $x = 16$ into $y = 4\sqrt{x} + 7$.

and $\frac{dy}{dx} = \frac{2}{\sqrt{16}} = \frac{1}{2}$ so gradient of normal $= -2$.

Gradient of normal $= -\frac{1}{\text{gradient of tangent}}$

Normal goes through (16, 23) with $m = -2$.

$y - 23 = -2(x - 16)$

Equation of normal is $y = -2x + 55$.

Example 17

The equation of a curve is $y = 4x - x^2$. The normal to the curve at the point $P(1, 3)$.
meets the curve again at Q. Find the coordinates of Q.

$y = 4x - x^2$

$\frac{dy}{dx} = 4 - 2x$

When $x = 1$, $\frac{dy}{dx} = 4 - 2 = 2$,

so gradient of the normal $= -\frac{1}{2}$.

Equation of the normal is $y - 3 = -\frac{1}{2}(x - 1)$.

So $y = -\frac{1}{2}x + \frac{7}{2}$

To find where the curve meets the normal
we solve $y = 4x - x^2$ and $y = -\frac{1}{2}x + \frac{7}{2}$ simultaneously.

$4x - x^2 = -\frac{1}{2}x + \frac{7}{2}$

$8x - 2x^2 = -x + 7$

Multiply both sides by 2.

$2x^2 - 9x + 7 = 0$

$(2x - 7)(x - 1) = 0$

$x = \frac{7}{2}, y = \frac{7}{4}$ or $x = 1, y = 3$

We already know it goes through (1, 3).

The normal meets the curve again at $\left(\frac{7}{2}, \frac{7}{4}\right)$.

g) Stationary points

Example 3

Find the coordinates of the stationary points of the curve $y = 6x - 2x^3$ and **determine** their nature. **Sketch** the curve.

$\frac{dy}{dx} = 6 - 6x^2 = 0$ at stationary points

Differentiate and put $\frac{dy}{dx}$ equal to 0.

$6 = 6x^2$

$x = 1$ and $x = -1$

Solve for x.

$y = 4$ and $y = -4$

Substitute x in to $y = 6x - 2x^3$ to find y.

x	-2	-1	0	1	2
$\frac{dy}{dx}$	-18	0	6	0	-18

Use values for x less than -1, between 0 and 1, and greater than 1.

Draw the positive and negative gradients.

(1, 4) is a maximum turning point.

(−1, −4) is a minimum turning point.

Write down if it is a maximum or minimum.

Note: $\frac{d^2y}{dx^2} = -12x$

This is an alternative method. It is useful when $\frac{dy}{dx}$ is simple to differentiate again.

When $x = 1$, $\frac{d^2y}{dx^2} = -12$, $\frac{d^2y}{dx^2} < 0$, so maximum.

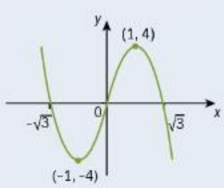
When $x = -1$, $\frac{d^2y}{dx^2} = 12$, $\frac{d^2y}{dx^2} > 0$, so minimum.

Sketch:

$y = 6x - 2x^3 = 2x(3 - x^2)$

Cuts x-axis at $0, \sqrt{3}, -\sqrt{3}$

Find when $y = 0$ for points where the graph crosses the x-axis.



Example 1

Find the values of x for which $f(x) = 5 - 4x - x^2$ is a decreasing function.

$f'(x) = -4 - 2x$ — First differentiate to give $f'(x)$.

Decreasing function so $-4 - 2x < 0$.

$-4 < 2x$ — Then solve for x in an inequality.

$x > -2$

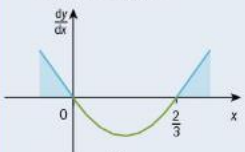
$f(x)$ is a decreasing function when $x > -2$.

Example 2

$y = x^3 - x^2$. Find the range of values of x for which y is increasing.

$\frac{dy}{dx} = 3x^2 - 2x$ — First differentiate to find $\frac{dy}{dx}$.

$= x(3x - 2) > 0$ — y is **increasing** so put $\frac{dy}{dx} > 0$.



Draw a diagram. Note that this is the graph of $\frac{dy}{dx}$.

$x < 0$ or $x > \frac{2}{3}$ — $\frac{dy}{dx} > 0$ so find where the graph lies **above** the x-axis, giving two intervals.

y is increasing when $x < 0$ and $x > \frac{2}{3}$.

Example 4

Find the coordinates of the stationary points of the curve $y = \frac{2}{2x-1} + x$ and determine their nature.

Let $A = \frac{2}{2x-1}$ and let $u = 2x - 1$

Use the chain rule to differentiate $\frac{2}{2x-1}$.

$\frac{du}{dx} = 2$

Let $A = \frac{2}{u} = 2u^{-1}$ $\quad \frac{dA}{du} = -2u^{-2} = \frac{-2}{(2x-1)^2}$

$\frac{dA}{dx} = \frac{-2}{(2x-1)^2}(2) = \frac{-4}{(2x-1)^2}$

$\frac{dy}{dx} = \frac{-4}{(2x-1)^2} + 1 = 0$ at stationary points

Put $\frac{dy}{dx} = 0$.

$(2x - 1)^2 = 4$

$4x^2 - 4x - 3 = 0$

Multiply each term by $(2x - 1)^2$ and rearrange.

$(2x + 1)(2x - 3) = 0$

$x = -\frac{1}{2}, y = -\frac{3}{2}$ or $x = \frac{3}{2}, y = \frac{5}{2}$

x	-1	$-\frac{1}{2}$	0	$\frac{3}{2}$	2
$\frac{dy}{dx}$	$\frac{5}{9}$	0	-3	0	$\frac{5}{9}$

Draw up a table to show the gradient of the tangent (or we could find $\frac{d^2y}{dx^2}$ etc.).

$\left(-\frac{1}{2}, -\frac{3}{2}\right)$ is a maximum turning point.

$\left(\frac{3}{2}, \frac{5}{2}\right)$ is a minimum turning point.

Draw the positive and negative gradients and determine the nature of the stationary points.

h) Rates of changes

Example 7

The radius, r, of a circle is increasing at the rate of $\frac{2}{r^2}\,\text{m}\,\text{s}^{-1}$.
Find the rate at which the area, A, is increasing when $r = 8$.

Follow the steps on the right of the examples for problems involving connected rates of change.

We want to find $\frac{dA}{dt}$.	Write down the rate of change we want to find.
$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$	Use the chain rule to write down an identity for $\frac{dA}{dt}$.
$\frac{dr}{dt} = \frac{2}{r^2}$	Write the information given as the rate of change of r with respect to t.
But, $A = \pi r^2$	Write down the link between A and r.
$\frac{dA}{dr} = 2\pi r$	Differentiate to the rate of change of A with respect to r.
So, $\frac{dA}{dt} = 2\pi r \times \frac{2}{r^2} = \frac{4\pi}{r}$	Substitute $\frac{dA}{dr}$ and $\frac{dr}{dt}$.
When $r = 8$, $\frac{dA}{dt} = \frac{4\pi}{8}$ $\frac{dA}{dt} = \frac{\pi}{2}$	We can now substitute $r = 8$.
$= 1.57\,\text{m}^2\,\text{s}^{-1}$ (3 s.f.)	Do not forget the units (area/time).

Example 8

A spherical balloon is being blown up so that its volume increases at a rate of $3\,\text{cm}^3\,\text{s}^{-1}$.
Find the rate of increase of the radius of the balloon when the volume of the balloon is $60\,\text{cm}^3$.

We want to find $\frac{dr}{dt}$.	Write down the rate of change we want to find.
$\frac{dr}{dt} = \frac{dr}{dV} \times \frac{dV}{dt}$	Write down the chain rule for $\frac{dr}{dt}$.
$\frac{dV}{dt} = 3$	Write the information given as a rate of change.
$V = \frac{4}{3}\pi r^3$, $\frac{dV}{dr} = 4\pi r^2$	Differentiate to get the rate of change of V with respect to r.
$\frac{dr}{dt} = \frac{1}{4\pi r^2} \times 3 = \frac{3}{4\pi r^2}$	$\frac{dr}{dV} = 1 \div \frac{dV}{dr}$
When $V = 60$, $60 = \frac{4}{3}\pi r^3$ $r^3 = \frac{45}{\pi}$ $r = 2.42859...$	Substitute for these rates of change.
$\frac{dr}{dt} = \frac{3}{4\pi r^2} = \frac{3}{4\pi(2.42859...)^2}$	We can now substitute for $r = 2.42859\ldots$
$= 0.0405\,\text{cm}\,\text{s}^{-1}$ (3 s.f.)	Do not forget the units (radius/time).

Summary:

Differentiating $y = x^n$

- If $y = x^n$ then $\frac{dy}{dx} = nx^{n-1}$

Differentiating $y = ax^n$

- If $y = ax^n$ then $\frac{dy}{dx} = anx^{n-1}$

We differentiate when

- we are given $f(x)$ and want to find $f'(x)$
- we are given y and are want to find $\frac{dy}{dx}$
- we want to find $\frac{d}{dx}(f(x))$
- we want to find the gradient of the tangent (or the normal) of a curve given the equation of the curve
- we want to find the coordinates of a point on a curve and are given the value of the gradient at that point
- we want to find the rate of change of y with respect to x given y.

The chain rule (differentiating a function of a function)

- The chain rule states that:
  If y is a function of u, and u is a function of x, $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

The second derivative

- If we differentiate $\frac{dy}{dx}$ we get $\frac{d^2y}{dx^2}$.
- If we differentiate $f'(x)$ we get $f''(x)$.

Increasing and decreasing functions

- When the gradient of a function is positive we say the function is an **increasing function**. A function $f(x)$ is increasing for $a < x < b$ if $f'(x) > 0$ for $a < x < b$.
- When the gradient of a function is negative we say the function is a **decreasing function**. A function $f(x)$ is decreasing for $a < x < b$ if $f'(x) < 0$ for $a < x < b$.

Stationary points

- Any point on a curve where $\frac{dy}{dx} = 0$ is called **a stationary point**.
- There are three types of stationary point: minimum points, maximum points and points of inflexion.
- At a minimum point, $\frac{d^2y}{dx^2} > 0$.
- At a maximum point, $\frac{d^2y}{dx^2} < 0$.
- At $\frac{d^2y}{dx^2} = 0$, the point may be a minimum, or a maximum, or a point of inflexion, and so we must look at the sign of $\frac{dy}{dx}$ either side of the stationary point to determine its nature.

Points of inflexion are not included in Cambridge International AS & A Level 9709 examination papers.

Problems involving maximum and minimum values

- We can use these steps to solve problems:
  - Draw a diagram where possible.
  - Choose letters to represent the unknown quantities.
  - Write an expression for the quantity we are trying to maximise/minimise – it may contain two unknowns.
  - Use a given condition from the question to write an equation connecting two variables.
  - Rearrange the equation to express one variable in terms of the other.
  - Use substitution to find an expression in terms of one variable only.
  - Use differentiation to find maximum/minimum values.

Connected rates of change

- Always derive a formula related to the rates of change given

  e.g. $\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$ e.g. $\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$

  e.g. $\frac{dV}{dt} = \frac{dV}{ds} \times \frac{ds}{dt}$ e.g. $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dt} \times \frac{dt}{dx}$.
- You can use these steps to solve problems:
  - Write the information given as a rate of change.
  - Write down the rate of change you want to find.
  - Write down the link between the two variables.
  - Differentiate to get another rate of change.
  - You may also need to use another formula and then differentiate this to get another rate of change.
  - Substitute these rates of change.
  - You can then substitute any given values to get the answer.